DECISIONTRACE · RIDDHIMA · OCTOBER 2026

When should you trust a simulated decision?

Evidence, uncertainty, and decision stability for AI simulations

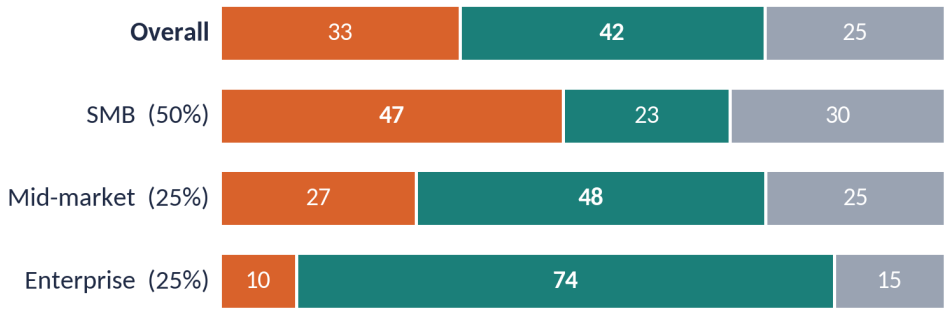
A simulation says **Option B is likely to outperform Option A.** A decision-maker's next questions are: **Why? How certain? Which populations drove it? What does it depend on? What would flip it?**

8,000 synthetic agents



each square ≈ 80 agents

Who prefers what (%)



42.1%
B Better reliability

33.0%
A Lower price

24.9%
C Easier setup

+9.0pp
B leads A

The headline hides two problems.

24.4pp

SMB is 50% of the population, and prefers A. There, A leads B by 24.4pp (47.3% vs 22.9%). The aggregate "B wins" is carried by Mid-market and Enterprise.

0.63

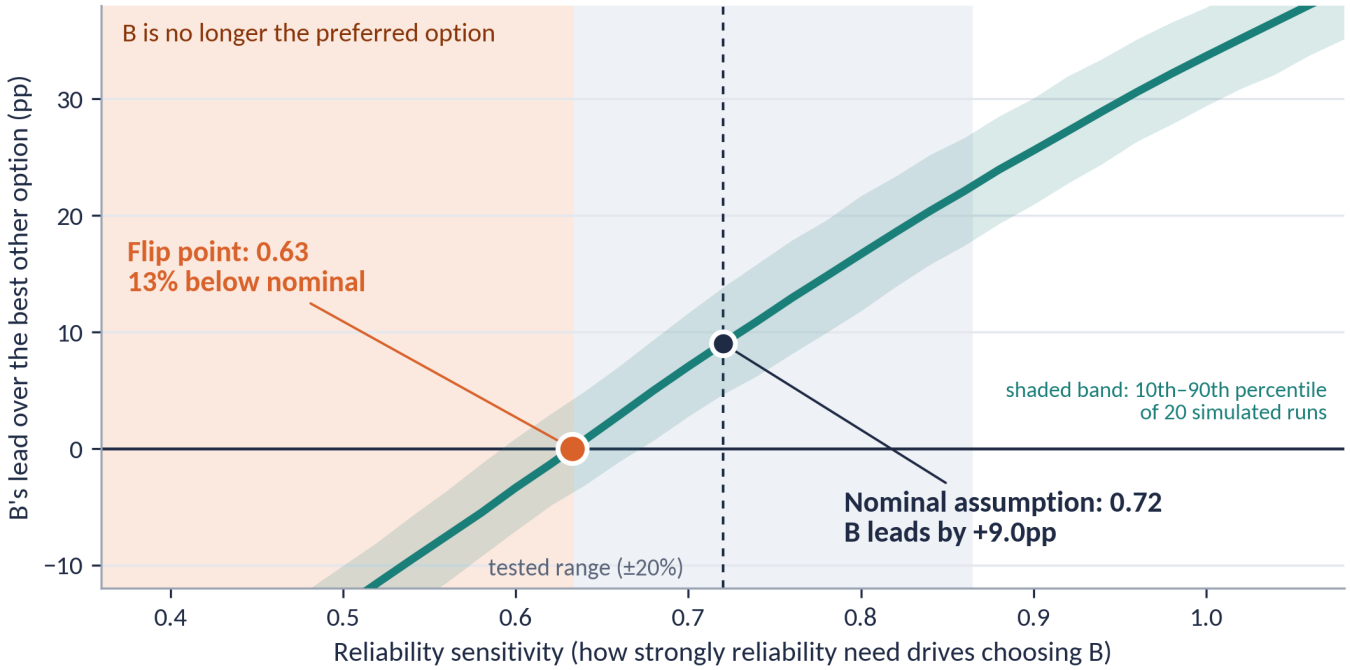
B's lead flips when reliability sensitivity falls below 0.63. The nominal assumption is 0.72. That is one unvalidated input, and a 13% change.

Simulation synthetic agents choose → **Evidence ledger** input · generated · derived → **Deterministic analysis** stability · segments · flips → **Decision presentation** changes wording with evidence

Scope. A synthetic test fixture. Not a model of real buyers. No real customer data. An independent experiment, not any company's models, data or methodology.

What would flip the decision?

Hold everything else at nominal and move one assumption: how much reliability matters to the simulated buyers.



Each point is the mean of the same 20 seeded runs used for the headline, so the nominal point matches page 1. The deterministic analysis in the repo puts the flip at 0.626.

Decision stability

74.8%

of 400 tested assumption sets keep B preferred

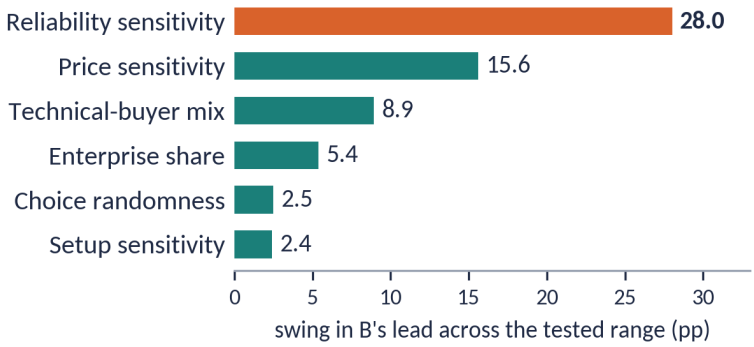


B preferred: 299

another option preferred: 101

Latin-hypercube samples; each assumption within its tested range.

What matters most?



The decision rests mostly on an unvalidated simulation input, not population mix.

Reliability sensitivity moves B's lead by 28pp across its tested range. Enterprise share moves it 5.4pp. A simulation can produce a decisive-looking answer while the decision itself stays fragile to assumptions nobody has validated. That is a product problem.

When should the product stop declaring a winner?

Seven controlled degradations of the same experiment. Each changes what the product is willing to say, even where the headline lead barely moves.

MISSING SEGMENT

INSUFFICIENT EVIDENCE

No Enterprise agents in the population.

A would win instead.

HIGH VARIANCE

UNSTABLE

Simulator noise ×11.

B is ahead in only 50% of runs. No robust preference.

SUBGROUP CONFLICT

CONTESTED

Aggregate favors B by 5.3pp.

SMB (58% of the population) prefers A by a wide margin.

STALE SIMULATION

OUT OF DATE

The run used input v1; the current input is v2.

Re-run before using.

POPULATION IMBALANCE
INSUFFICIENT EVIDENCE
Enterprise is 25% of the result from 169 agents.

INSUFFICIENT SAMPLE
INSUFFICIENT EVIDENCE
Only 300 agents (minimum 1,000).

NEAR TIE
UNSTABLE
B ahead by 2.3pp, in 65% of runs.

Weak-evidence states never claim a leader.

A NAIVE PRODUCT SAYS

B wins.

Same run. Same data.
No uncertainty, no dissent, no assumptions shown.

DECISIONTRACE SAYS, SAME RUN

MODERATE · CONFIDENCE: MODERATE
B (Better reliability) leads A by 9.0pp in the simulation, but the lead depends on assumptions.
Decision stability: **MODERATE** · B remains preferred under **75%** of 400 tested assumption sets · leads in 100% of runs.

The aggregate result masks significant population disagreement. SMB prefers A, 24.4pp ahead of B. **Most sensitive assumption:** reliability sensitivity.

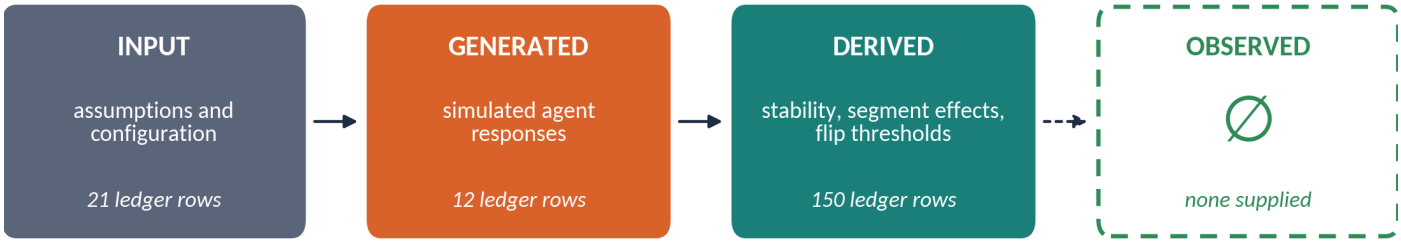
AND WHEN THE EVIDENCE WEAKENS: A NEAR TIE, B AHEAD BY ONLY 2.3pp

UNSTABLE · CONFIDENCE: INSUFFICIENT
Result unstable: insufficient evidence for a robust preference.
Why: B is not ahead in every plausible run (worst-case gap -3.0pp); B is ahead in only 65% of runs; near tie: the gap is 2.3pp.

Don’t let explanation become evidence.

A follow-up answer can only cite the ledger, and the ledger knows what kind of evidence each row is.

A USER ASKS	THE SYSTEM ANSWERS
Did real customers prefer B?	**The simulation does not establish that.** Every preference figure here is generated by simulated agents, and no observed customer outcome data was supplied `[EV-0005]`.



Every number in an answer links to a ledger row and its provenance chain. The overall lead, for example, traces back through: `overall.lead` **DERIVED** ← `overall.share.A, overall.share.B` **DERIVED** ← `generated.runs.overall` **GENERATED** ← `input.n_agents, input.n_runs` **INPUT**.

Then I tried to break the evaluator.

408/408

grounded answers pass all five checks

citation · numerical fidelity · observed vs generated · uncertainty · overreach

2,698/2,698

deliberately corrupted answers caught

dropped citations, altered numbers, added certainty, invented causes

1/7

subtle, novel overreach answers caught

0 of 6 clean paraphrases flagged. High precision, low recall.

The responder and the checks were built together, so the 100% measures consistency, not generalization. The corrupted answers use the phrases the checks were written for. The seven subtle ones do not.

Provenance can constrain what an explanation is allowed to claim. But deterministic semantic guardrails do not generalize well enough on their own.

What I’d test next

Real observed outcomes
Compare the simulation with reality.

Calibration
Does reported stability match actual decision robustness?

LLM judge + human audit
Catch semantic overreach the lexicons miss.

Other population models
How much does DecisionTrace itself depend on the simulator?

Independent synthetic experiment. Not any company’s models, data, APIs or methodology.